Evaluation runs: **ring-el** (ring-election-igloo)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	4	202	conforms	0.0	4	1 h, 250 GB	4	10	brain03, 2026-11-03
3	12	7506	conforms	0.2	4	1 h, 250 GB	12	78	brain03, 2026-11-03
4	48	343704	conforms	4.5	4	1 h, 250 GB	48	1176	brain03, 2026-11-03
5	240	19172520	conforms	178.0	42	1 h, 250 GB	240	28920	brain03, 2026-11-03
6	1440	1271837520	conforms	6253.0	2168	48 h, 250 GB	1440	1037520	brain03, 2026-11-07

stateless: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	4	194	24	violation	0.0	2	1 h, 250 GB	28	106	brain03, 2026-11-03
3	12	7314	576	violation	0.3	2	1 h, 250 GB	588	6990	brain03, 2026-11-03
4	48	339096	13824	violation	7.6	2	1 h, 250 GB	13872	664728	brain03, 2026-11-03
5	240	19049640	368640	violation	246.6	48	1 h, 250 GB	368880	88502520	brain03, 2026-11-03
6	1440	1268151120	11059200	violation	25099.0	2196	48 h, 250 GB	11060640	15926285520	brain03, 2026-11-08

Not run (stateless). left out by the table rules (timeout): 6 correct bag, 6 no-compare bag.

skipped: a smaller size had reached its limit: 7 no-compare bag, 7 correct bag, 8 no-compare bag, 8 correct bag, 9 no-compare bag, 9 correct bag, 10 no-compare bag, 10 correct bag.

planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 no-compare bag, 7 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	4	202	conforms	0.8	2	1 h, 250 GB	brain02, 2026-11-03
3	12	7506	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-03
4	48	343704	conforms	4.8	2	1 h, 250 GB	brain02, 2026-11-03
5	240	19172520	conforms	96.4	44	48 h, 250 GB	brain05, 2026-11-02
6	1440	1271837520	conforms	6215.0	2172	48 h, 250 GB	brain03, 2026-11-07

stateful: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	4	194	24	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	12	7314	576	violation	0.2	2	1 h, 250 GB	brain02, 2026-11-03
4	48	339096	13824	violation	4.8	2	1 h, 250 GB	brain02, 2026-11-03
5	240	19049640	368640	violation	101.3	48	48 h, 250 GB	brain05, 2026-11-02
6	1440	1268151120	11059200	violation	6516.0	2186	48 h, 250 GB	brain03, 2026-11-08
7				(48 h)			48 h, 250 GB	brain02, 2026-11-05
8				(48 h)			48 h, 250 GB	brain04, 2026-11-03

Not run (stateful). left out by the table rules (ok): 5 correct bag, 5 no-compare bag, 6 correct bag, 6 no-compare bag.

left out by the table rules (timeout): 6 correct bag, 6 no-compare bag.

skipped: a smaller size had reached its limit: 7 no-compare bag, 7 correct bag, 8 no-compare bag, 8 correct bag, 9 no-compare bag, 9 correct bag, 10 no-compare bag, 10 correct bag.

the tool ended with an error (exit-143): 7 correct bag.

planned but never scheduled (grid points beyond the ladder): 8 no-compare bag, 8 correct bag, 9 correct bag, 10 no-compare bag, 10 correct bag, 11 no-compare bag, 11 correct bag, 12 correct bag, 12 no-compare bag, 13 correct bag, 13 no-compare bag.

left out by the table rules (invalid: foreign load): 8 correct bag, 9 no-compare bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	202	conforms	0.1	4	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03
3	12	7506	conforms	0.4	6	4 h, 250 GB	0.0	0.4	0.0	merge	brain02, 2026-11-03
4	48	343704	conforms	4.9	6	4 h, 250 GB	0.2	4.9	0.0	merge	brain02, 2026-11-03
5	240	19172520	conforms	174.0	40	4 h, 250 GB	0.2	174.0	0.0	merge	brain02, 2026-11-03
6	1440	1271837520	conforms	6055.0	2138	48 h, 250 GB	0.1	6055.0	0.0	merge	brain03, 2026-11-08

offline: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	194	2	violation	0.1	6	4 h, 250 GB	0.1	0.1	0.0	merge	brain02, 2026-11-03
3	12	7314	12	violation	0.4	8	4 h, 250 GB	0.0	0.4	0.0	merge	brain02, 2026-11-03
4	48	339096	72	violation	5.0	6	4 h, 250 GB	0.2	5.0	0.0	merge	brain02, 2026-11-03
5	240	19049640	480	violation	264.8	40	4 h, 250 GB	0.2	264.8	0.0	merge	brain02, 2026-11-03
6	1440	1268151120	3600	violation	6081.1	2132	48 h, 250 GB	0.1	6081.0	0.1	merge	brain03, 2026-11-08

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 no-compare bag.

skipped: a smaller size had reached its limit: 6 no-compare bag, 6 correct bag, 7 no-compare bag, 7 correct bag, 8 no-compare bag, 8 correct bag, 9 no-compare bag, 9 correct bag, 10 no-compare bag, 10 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	202	conforms	5.3	156	4 h, 250 GB	0.1	5.3	0.0	merge	brain02, 2026-11-03
3	12	7506	conforms	13.7	592	10 min, 16 GB	0.0	13.7	0.0	in-memory	brain01, 2026-11-02
4	48	343704	conforms	505.1	1752	10 min, 16 GB	0.1	505.0	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	194	2	violation	4.5	162	4 h, 250 GB	0.1	4.5	0.0	merge	brain02, 2026-11-03
3	12	7314	12	violation	12.9	534	10 min, 16 GB	0.0	12.8	0.0	in-memory	brain01, 2026-11-02
4	48	339096	72	violation	487.9	1485	10 min, 16 GB	0.1	487.8	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 correct bag.
skipped: a smaller size had reached its limit: 7 no-compare bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	202	conforms	4.2	166	4 h, 250 GB	4.2	0.1	0.0	merge	brain02, 2026-11-03
3	12	7506	conforms	9.6	150	4 h, 250 GB	9.6	0.4	0.0	merge	brain02, 2026-11-03
4	48	343704	conforms	4.9	163	4 h, 250 GB	4.1	4.9	0.0	merge	brain02, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	194	2	violation	4.2	168	4 h, 250 GB	4.2	0.1	0.0	merge	brain02, 2026-11-03
3	12	7314	12	violation	9.6	152	4 h, 250 GB	9.6	0.4	0.0	merge	brain02, 2026-11-03
4	48	339096	72	violation	5.0	163	4 h, 250 GB	4.1	5.0	0.0	merge	brain02, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 no-compare bag.
skipped: a smaller size had reached its limit: 7 correct bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	202	conforms	1.1	152	1 h, 250 GB		brain05, 2026-11-02
3	12	7506	conforms	12.8	458	1 h, 250 GB		brain05, 2026-11-02
4	48	343704	conforms	652.9	1593	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	194	24	violation	1.1	150	1 h, 250 GB	99	brain05, 2026-11-02
3	12	7314	576	violation	12.2	403	1 h, 250 GB	333	brain05, 2026-11-02
4	48	339096	13824	violation	647.9	1490	1 h, 250 GB	6726	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 correct bag, 2 no-compare bag, 3 correct bag, 3 no-compare bag, 4 no-compare bag, 4 correct bag, 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 correct bag.

left out by the table rules (invalid: killed after 26536 s of 172800): 7 no-compare bag.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	202	conforms	3.1	159	1 h, 250 GB		brain04, 2026-11-01
3	12	7506	conforms	3.3	152	1 h, 250 GB		brain04, 2026-11-01
4	48	343704	conforms	7.7	164	1 h, 250 GB		brain04, 2026-11-01
7	10080		(48 h)			48 h, 80 GB		pinky10, 2026-11-05

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-compare**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	194	24	violation	3.1	159	1 h, 250 GB	3	brain04, 2026-11-01
3	12	7314	576	violation	3.3	152	1 h, 250 GB	12	brain04, 2026-11-01
4	48	339096	13824	violation	7.6	164	1 h, 250 GB	19	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 5 no-compare bag, 6 no-compare bag, 6 correct bag, 7 no-compare bag.

left out by the table rules (ok): 2 correct bag, 2 no-compare bag, 3 correct bag, 3 no-compare bag, 4 no-compare bag, 4 correct bag.